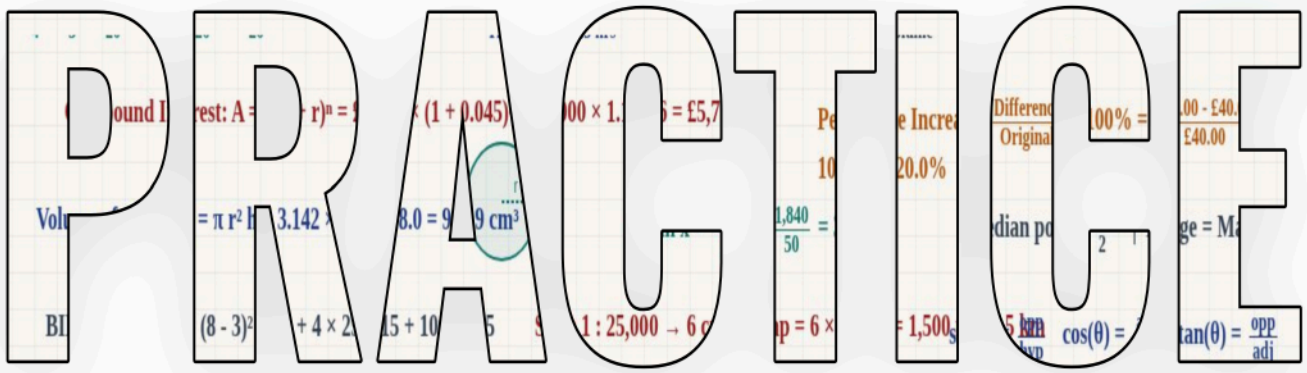
MASTER FUNCTIONAL SKILLS

FUNCTIONAL SKILLS MATHEMATICS

COMPANION PRACTICE WORKBOOK • STEPPED DRILLS

LEVEL 1



PRACTICE COMPANION

Guided calculation frames & real-life workplace maths scenarios

Scaffolded Grids | Real Scenarios | Step-by-Step | Answer Key Included

MASTER FUNCTIONAL SKILLS • REFORMED MATHEMATICS CURRICULUM DIRECTORY

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34 Guided Workplace Practice Drills, 5mm Working Grids, Case Studies & 2 Full Practice Papers

PEARSON EDEXCEL MATHS SUITE

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Master Functional Skills • Mathematics Manual • Number, Measures, Shape, Space & Data Handling **Lead Tutor:** Abdul • **Edition:** 2026/2027 Reformed Standards

DRILL 1 OF 34 • LEVEL 1 STANDARDS

1 Worksheet 1: Place Value & Negative Temperatures

Worksheet 1: Place Value & Negative Temperatures: Everyday...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Warm-Up Drills:** a) Write in figures: 'Forty-two thousand, five hundred and eight' b) Write in words: 306,420. c) Put in order, coldest first: +3°C, -6°C, -14°C, +1°C, -2°C. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Warehouse Cold Store Monitoring:** A cold storage room is at **-16°C**. A refrigeration fault causes the temperature to rise by **9°C**. What is the new temperature? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 1: Place Value & Negative Temperatures • Candidate Workspace

✎ Show all working for Worksheet 1 tasks below. Check your final answers.

DRILL 2 OF 34 • LEVEL 1 STANDARDS

2 Worksheet 2: Column Addition & Subtraction

Worksheet 2: Column Addition & Subtraction: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Column Addition & Subtraction Warm-Up:** Calculate without a calculator: a) 1,458 + 2,385 b) 4,000 - 1,675. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Hospital Supply Delivery:** A dispensary receives three boxes of medical masks: 450 in box A, 625 in box B, and 380 in box C. How many masks are there in total? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 2: Column Addition & Subtraction • Candidate Workspace

✎ Show all working for Worksheet 2 tasks below. Check your final answers.

3

DRILL 3 OF 34 • LEVEL 1 STANDARDS

Worksheet 3: Multiplication & Division

Worksheet 3: Multiplication & Division: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Written Calculation Warm-Up:** Calculate: a) 236 × 18 b) 952 ÷ 7. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Packaging Pallet Quantities:** A food factory packs 24 tins of soup per cardboard carton. A supermarket orders 35 cartons. How many tins is this in total? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 3: Multiplication & Division • Candidate Workspace

✎ Show all working for Worksheet 3 tasks below. Check your final answers.

DRILL 4 OF 34 • LEVEL 1 STANDARDS

4 Worksheet 4: Multiplying & Dividing by 10, 100

Worksheet 4: Multiplying & Dividing by 10, 100: Everyday Con...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Powers of 10 Warm-Up:** Calculate: a) 34 × 100 b) 5.8 × 10 c) 4,200 ÷ 100 d) 65 ÷ 10. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Bulk Stationery Purchase:** A school buys 100 packs of dry-wipe markers at £3.45 per pack. How much does the school pay in total? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 4: Multiplying & Dividing by 10, 100 • Candidate Workspace

✎ Show all working for Worksheet 4 tasks below. Check your final answers.

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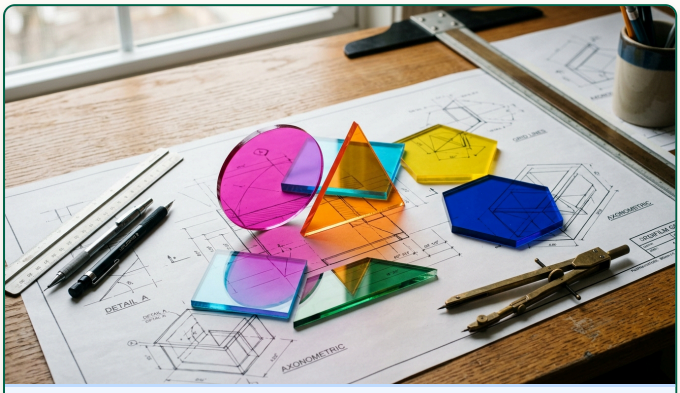
DRILL 5 OF 34 • LEVEL 1 STANDARDS

Worksheet 5: Order of Operations (BIDMAS) & Rules



Worksheet 5: Order of Operations (BIDMAS) & Rules: Everyda...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: BIDMAS Warm-Up Drills:** Evaluate: a) 14 + 3 × 5 b) (18 - 6) ÷ 3 c) 25 - 4 × 2 + 1 d) 4^2 - 10. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Equipment Hire Formula:** Tool hire cost: Cost = £30 + (£15 × days). Calculate the total hire cost for 4 days. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

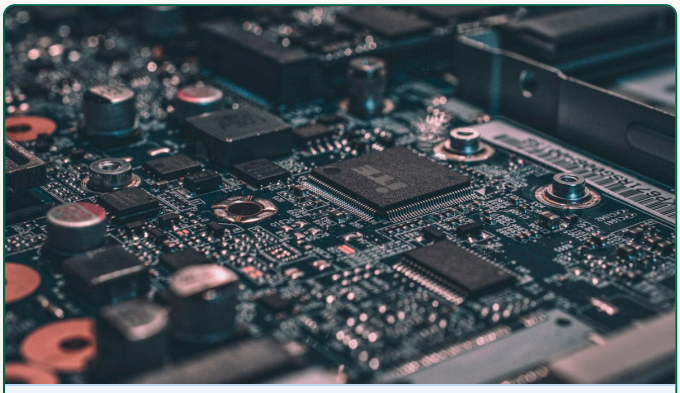
Worksheet 5: Order of Operations (BIDMAS) & Rules • Candidate Workspace

✎ Show all working for Worksheet 5 tasks below. Check your final answers.

6

DRILL 6 OF 34 • LEVEL 1 STANDARDS

Worksheet 6: Simplifying & Equivalent Fractions



Worksheet 6: Simplifying & Equivalent Fractions: Everyday Co...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Simplifying Fractions Warm-Up:** Simplify fully: a) 14/21 **(4 marks)**
b) 20/50 c) 18/24 d) 15/45.

Show your working

Answer: _____

2 **Task 2: Equivalence Grid Missing Values:** Find missing values: a) **(3 marks)**
2/3 = ? ÷ 15 b) 5/8 = 20 ÷ ? c) ? ÷ 10 = 12/20.

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 6: Simplifying & Equivalent Fractions • Candidate Workspace

✎ Show all working for Worksheet 6 tasks below. Check your final answers.

DRILL 7 OF 34 • LEVEL 1 STANDARDS

7 Worksheet 7: Fractions of Amounts

Worksheet 7: Fractions of Amounts: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Rapid Fraction Calculations:** Calculate: a) 1/3 of £45 b) 1/5 of 80 kg c) 3/4 of £60 d) 2/7 of 35 m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Pharmacy Pill Inventory:** A pharmacy technician has 240 antibiotic capsules. In one morning, 3/8 are dispensed. How many capsules remain? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 7: Fractions of Amounts • Candidate Workspace

✎ Show all working for Worksheet 7 tasks below. Check your final answers.

DRILL 8 OF 34 • LEVEL 1 STANDARDS

8 Worksheet 8: Decimals & Cafe Receipt

Worksheet 8: Decimals & Cafe Receipt: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Decimal Arithmetic Drills:** Calculate: a) 15.45 + 8.90 b) 50 - 24.35 c) 6.4 × 5 d) 18.60 ÷ 3. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Cafe Till Receipt:** A customer buys: 2 coffees at £2.85 each, 1 sandwich for £4.50, and 1 muffin for £2.20. How much change from a £20 note? **(3 marks)**

Show your working

Answer: _____

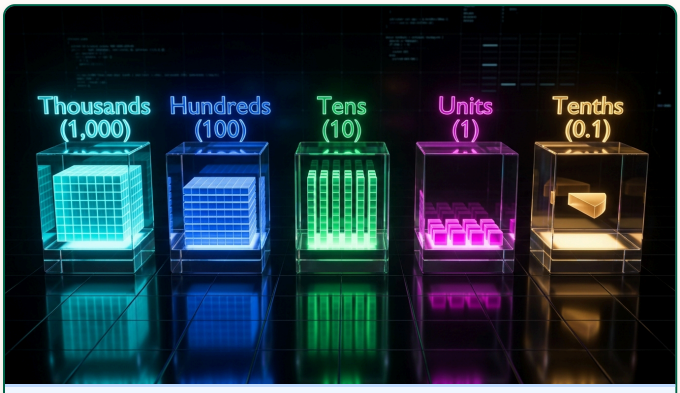
5MM RULED GRID WORKING AREA

Worksheet 8: Decimals & Cafe Receipt • Candidate Workspace

✎ Show all working for Worksheet 8 tasks below. Check your final answers.

DRILL 9 OF 34 • LEVEL 1 STANDARDS

9 Worksheet 9: Benchmark Percentages & Discounts



Worksheet 9: Benchmark Percentages & Discounts: Everyday ...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Benchmark Percentage Drills:** Find: a) 50% of £84 b) 25% of 60 kg c) 10% of £350 d) 5% of £180. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Retail Clearance Discount:** A mountain bike originally costs £320.00. It is discounted by 25% in an autumn sale. What is the new sale price? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 9: Benchmark Percentages & Discounts • Candidate Workspace

✎ Show all working for Worksheet 9 tasks below. Check your final answers.

DRILL 10 OF 34 • LEVEL 1 STANDARDS

10 Worksheet 10: Fraction, Decimal & Percentage Equivalence

Worksheet 10: Fraction, Decimal & Percentage Equivalence: E...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Complete Equivalence Table:** Fill in blanks: a) Fraction 1/2 = Decimal _____ = _____ % b) Decimal 0.25 = Fraction _____ = _____ % **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Comparing Different Formats:** Which is larger: **3/5** or **65%**? Convert both to percentages or decimals to prove your answer. **(3 marks)**

Show your working

Answer: _____

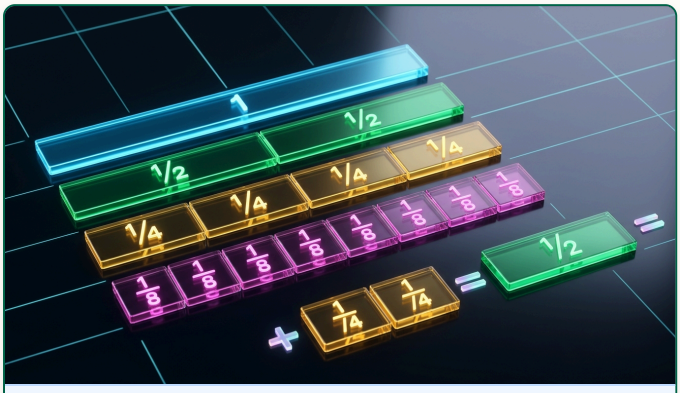
5MM RULED GRID WORKING AREA

Worksheet 10: Fraction, Decimal & Percentage Equivalence • Candidate Workspace

✎ Show all working for Worksheet 10 tasks below. Check your final answers.

DRILL 11 OF 34 • LEVEL 1 STANDARDS

11 Worksheet 11: Ratio Sharing & Punch Recipe



Worksheet 11: Ratio Sharing & Punch Recipe: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Ratio Sharing Warm-Up:** a) Share £60 in ratio 2 : 1. b) Share 40 kg in ratio 3 : 5. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Fruit Punch Recipe:** A party punch mixes orange juice and lemonade in ratio 3 : 2. An event host makes 15 litres of punch. How much orange juice is needed? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 11: Ratio Sharing & Punch Recipe • Candidate Workspace

✎ Show all working for Worksheet 11 tasks below. Check your final answers.

DRILL 12 OF 34 • LEVEL 1 STANDARDS

12 Worksheet 12: Metric Length Conversions

Worksheet 12: Metric Length Conversions: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Length Conversion Warm-Up:** Convert: a) 4.5 m to cm b) 250 cm to m c) 65 mm to cm d) 3.2 km to m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Joinery Skirting Trim:** A joiner has a 3 m strip of timber trim. He cuts three pieces: 85 cm, 90 cm, and 65 cm. How many cm remain? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

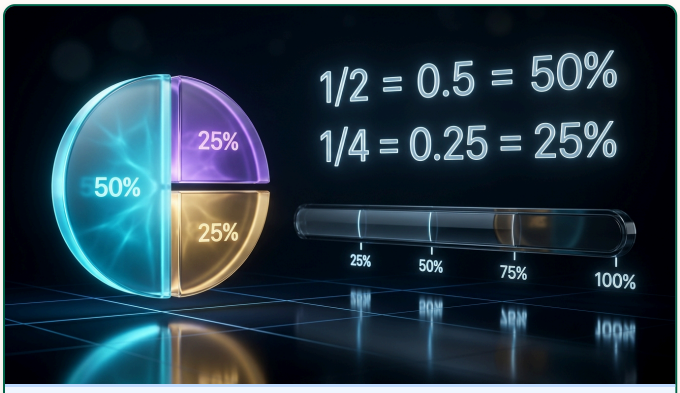
Worksheet 12: Metric Length Conversions • Candidate Workspace

✎ Show all working for Worksheet 12 tasks below. Check your final answers.

DRILL 13 OF 34 • LEVEL 1 STANDARDS

13

Worksheet 13: Metric Weight & Capacity



Worksheet 13: Metric Weight & Capacity: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Weight & Capacity Warm-Up:** Convert: a) 3.2 kg to grams b) 750 ml to litres c) 4,500 g to kg d) 1.5 L to ml. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Bakery Ingredient Mass:** A baker has 5 kg of sugar. He uses 1,250 g for bread and 850 g for pastries. How many grams of sugar remain? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 13: Metric Weight & Capacity • Candidate Workspace

✎ Show all working for Worksheet 13 tasks below. Check your final answers.

DRILL 14 OF 34 • LEVEL 1 STANDARDS

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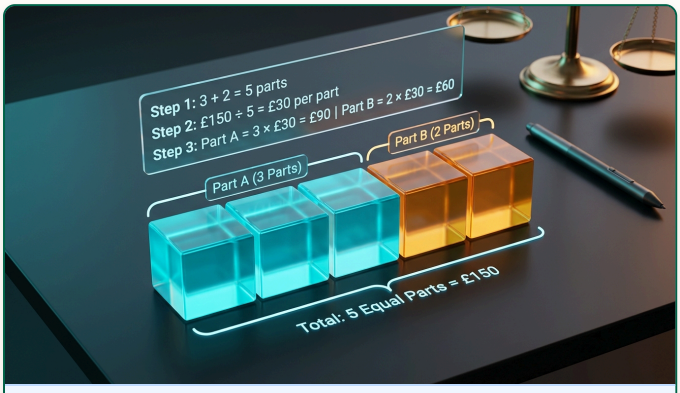
Worksheet 14: Imperial & Metric Approximations



Worksheet 14: Imperial & Metric Approximations: Everyday C...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Imperial Conversion Warm-Up:** Using standard approximations: a) 25 miles to km b) 40 km to miles c) 10 inches to cm. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Luggage Weight Allowance:** An airline allows 44 lbs of baggage. A passenger's suitcase weighs 18 kg. Using 1 kg = 2.2 lbs, is it within limit? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 14: Imperial & Metric Approximations • Candidate Workspace

✎ Show all working for Worksheet 14 tasks below. Check your final answers.

DRILL 15 OF 34 • LEVEL 1 STANDARDS

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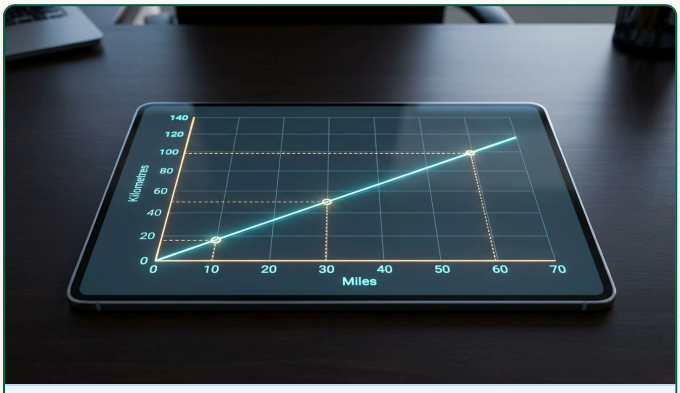
Worksheet 15: 24-Hour Time & Journey Duration



Worksheet 15: 24-Hour Time & Journey Duration: Everyday C...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Time Conversion Drills:** Convert to 24-hour format: a) 8:40 am b) 3:25 pm c) 9:10 pm d) 11:55 pm. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Train Commute Duration:** A commuter boards a train at 07:42. The train arrives at destination at 09:15. How long was the journey? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 15: 24-Hour Time & Journey Duration • Candidate Workspace

✎ Show all working for Worksheet 15 tasks below. Check your final answers.

DRILL 16 OF 34 • LEVEL 1 STANDARDS

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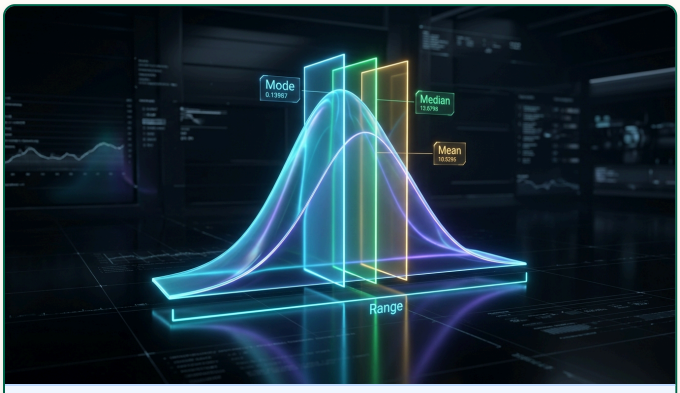
Worksheet 16: Perimeter & Skirting Boards



Worksheet 16: Perimeter & Skirting Boards: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Perimeter Warm-Up:** Find perimeter: a) Rectangle 8 m × 5 m b) Square of side 7.5 cm c) Equilateral triangle of side 9 cm. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Skirting Board with Doorway:** A living room is 5.5 m long and 4.0 m wide. There is a doorway opening of 1.0 m where no skirting is fitted. Find skirting length. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 16: Perimeter & Skirting Boards • Candidate Workspace

✎ Show all working for Worksheet 16 tasks below. Check your final answers.

DRILL 17 OF 34 • LEVEL 1 STANDARDS

17

Worksheet 17: Area of Rectangles & Carpet Cost

Worksheet 17: Area of Rectangles & Carpet Cost: Everyday Co...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Area Formula Warm-Up:** Find area: a) Rectangle 7 m × 4 m b) Square with side 9 cm c) Classroom 12 m × 8 m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Bedroom Carpet Cost:** A bedroom measures 4.5 m by 3.0 m. Carpet costs £16.00 per square metre. How much does it cost to carpet the room? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 17: Area of Rectangles & Carpet Cost • Candidate Workspace

✎ Show all working for Worksheet 17 tasks below. Check your final answers.

DRILL 18 OF 34 • LEVEL 1 STANDARDS

18 Worksheet 18: Composite L-Shape Area

Worksheet 18: Composite L-Shape Area: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Split & Sum Warm-Up:** An L-shape splits into: Rectangle A (6 m × 4 m) and Rectangle B (3 m × 2 m). Find total area. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Hallway Vinyl Flooring:** An L-shaped hallway has outer dimensions 8 m and 5 m, with cutout 4 m × 3 m. Find floor area. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

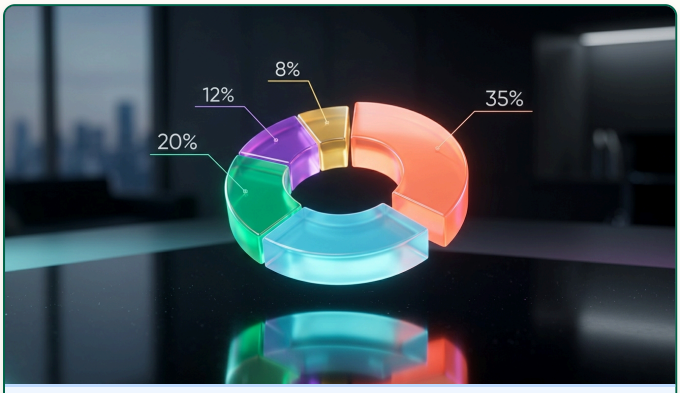
Worksheet 18: Composite L-Shape Area • Candidate Workspace

✎ Show all working for Worksheet 18 tasks below. Check your final answers.

DRILL 19 OF 34 • LEVEL 1 STANDARDS

19

Worksheet 19: Volume of Cuboids & Aquariums



Worksheet 19: Volume of Cuboids & Aquariums: Everyday Con...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Volume Calculation Warm-Up:** Calculate volume: a) Cuboid 5 cm × 4 cm × 10 cm b) Container 2 m × 1.5 m × 0.8 m. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Fish Aquarium Capacity:** An aquarium is 60 cm long, 30 cm wide, and 40 cm deep. How many litres of water does it hold? **(3 marks)**

Show your working

Answer: _____

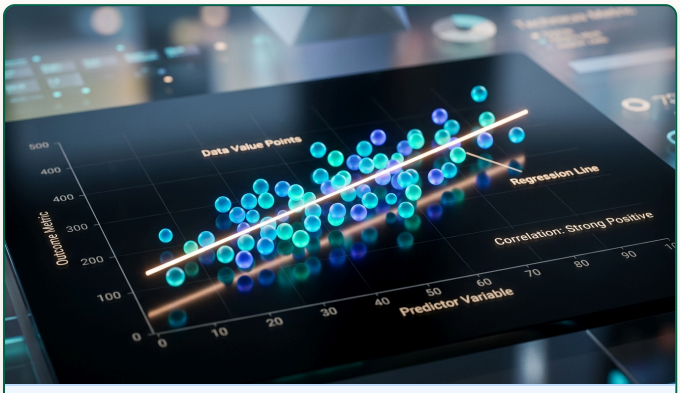
5MM RULED GRID WORKING AREA

Worksheet 19: Volume of Cuboids & Aquariums • Candidate Workspace

✎ Show all working for Worksheet 19 tasks below. Check your final answers.

DRILL 20 OF 34 • LEVEL 1 STANDARDS

20 Worksheet 20: Angles & Triangles



Worksheet 20: Angles & Triangles: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Classify Angles:** Classify: a) 35° b) 90° c) 125° d) 180°. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Missing Angle in Triangle:** A right-angled triangle has one acute angle measuring 38°. Calculate the size of the other acute angle. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 20: Angles & Triangles • Candidate Workspace

✎ Show all working for Worksheet 20 tasks below. Check your final answers.

DRILL 21 OF 34 • LEVEL 1 STANDARDS

21 Worksheet 21: 3D Shapes & Packaging Nets

Worksheet 21: 3D Shapes & Packaging Nets: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: 3D Properties Warm-Up:** State faces, edges, vertices: a) Cube b) Triangular prism c) Square-based pyramid. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Cardboard Packaging Net:** A package is made from a net with 6 rectangular faces arranged in 3 matching pairs. Name the 3D solid. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 21: 3D Shapes & Packaging Nets • Candidate Workspace

✎ Show all working for Worksheet 21 tasks below. Check your final answers.

DRILL 22 OF 34 • LEVEL 1 STANDARDS

22 Worksheet 22: Scale Plans & Real Dimensions

Worksheet 22: Scale Plans & Real Dimensions: Everyday Conte...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Scale Multipliers Warm-Up:** Scale 1 cm = 3 m: a) 4 cm on map = real length? b) 18 m real = map length in cm? **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Classroom Scale Drawing:** On a plan drawn to 1 : 50, a classroom is 14 cm long and 10 cm wide. What are its real dimensions in metres? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 22: Scale Plans & Real Dimensions • Candidate Workspace

✎ Show all working for Worksheet 22 tasks below. Check your final answers.

DRILL 23 OF 34 • LEVEL 1 STANDARDS

23 Worksheet 23: Compass Directions & Routes

Worksheet 23: Compass Directions & Routes: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Compass Direction Drills:** State direction: a) Opposite of North-East b) Facing North, turn 90° clockwise c) Facing East, turn 180°. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Delivery Van Route:** A delivery van leaves the depot travelling East for 6 miles, then turns South for 8 miles. In what direction is depot from the van? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 23: Compass Directions & Routes • Candidate Workspace

✎ Show all working for Worksheet 23 tasks below. Check your final answers.

DRILL 24 OF 34 • LEVEL 1 STANDARDS

24 Worksheet 24: Average Speed Calculations

Worksheet 24: Average Speed Calculations: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Speed Formula Warm-Up:** Find speed: a) 120 miles in 2 hours b) 90 miles in 1.5 hours c) 150 km in 3 hours. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Courier Van Journey:** A courier drives 165 miles in 3 hours. Calculate her average speed in miles per hour. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 24: Average Speed Calculations • Candidate Workspace

✎ Show all working for Worksheet 24 tasks below. Check your final answers.

DRILL 25 OF 34 • LEVEL 1 STANDARDS

25 Worksheet 25: Tally Tables & Fractions of Totals

Worksheet 25: Tally Tables & Fractions of Totals: Everyday Co...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Complete Tally Table:** 30 customers surveyed: Red (12), Blue (11), Green (7). Write tally marks and verify the total. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Clinic Appointment Audit:** A dental practice logs attendance: Attended (45), Cancelled (10), No-show (5). What fraction of appointments were attended? Simplify. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 25: Tally Tables & Fractions of Totals • Candidate Workspace

✎ Show all working for Worksheet 25 tasks below. Check your final answers.

DRILL 26 OF 34 • LEVEL 1 STANDARDS

26 Worksheet 26: Reading Bar Charts

Worksheet 26: Reading Bar Charts: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Read Bar Chart Warm-Up:** Bar heights: Dept A 40, Dept B 65, Dept C 55, Dept D 80. Identify: a) Highest dept b) Total across all 4 depts. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Difference Between Categories:** From Task 1, how many more staff are in Dept D compared to Dept A? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 26: Reading Bar Charts • Candidate Workspace

✎ Show all working for Worksheet 26 tasks below. Check your final answers.

DRILL 27 OF 34 • LEVEL 1 STANDARDS

27

Worksheet 27: Dual Bar Charts & Comparisons

Worksheet 27: Dual Bar Charts & Comparisons: Everyday Cont...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Dual Chart Reading:** Sales (Week 1 vs Week 2): Mon (30 vs 40), Tue (25 vs 35), Wed (45 vs 50). Find total sales in Week 2. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Day of Greatest Improvement:** On which day did sales increase the most between Week 1 and Week 2? What was the increase? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

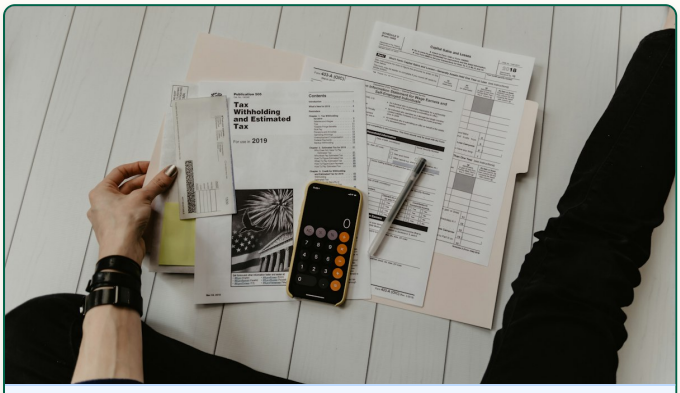
Worksheet 27: Dual Bar Charts & Comparisons • Candidate Workspace

✎ Show all working for Worksheet 27 tasks below. Check your final answers.

DRILL 28 OF 34 • LEVEL 1 STANDARDS

28

Worksheet 28: Pictograms & Key Multipliers



Worksheet 28: Pictograms & Key Multipliers: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Pictogram Reading Drills:** Key: 1 circle = 10 items. Find items for: a) 3 circles b) 4 1/2 circles c) 2 1/4 circles. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Library Book Borrowing:** Key: 1 star = 8 books borrowed. Friday shows 4 stars. Saturday shows 5 1/2 stars. Total borrowed over both days? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 28: Pictograms & Key Multipliers • Candidate Workspace

✎ Show all working for Worksheet 28 tasks below. Check your final answers.

DRILL 29 OF 34 • LEVEL 1 STANDARDS

29 Worksheet 29: Line Graphs & Trends

Worksheet 29: Line Graphs & Trends: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Temperature Graph Reading:** Temperatures: 08:00 (14°C), 10:00 (18°C), 12:00 (22°C). Estimate temp at 09:00 and 11:00. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Daily Sales Trend:** A coffee stall records daily takings: Mon £120, Tue £150, Wed £140, Thu £180, Fri £210. Describe the overall trend. **(3 marks)**

Show your working

Answer: _____

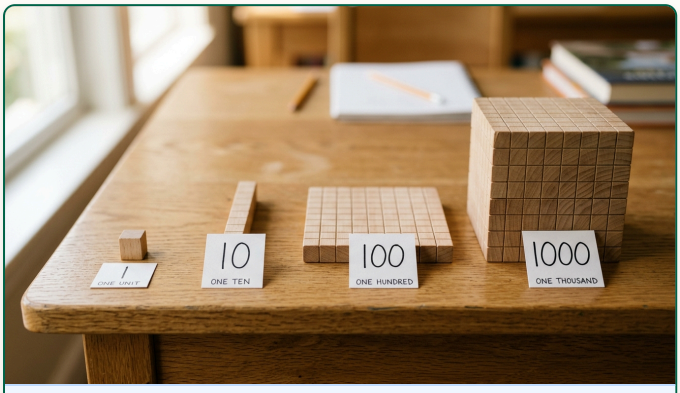
5MM RULED GRID WORKING AREA

Worksheet 29: Line Graphs & Trends • Candidate Workspace

✎ Show all working for Worksheet 29 tasks below. Check your final answers.

DRILL 30 OF 34 • LEVEL 1 STANDARDS

30 Worksheet 30: Pie Chart Proportions



Worksheet 30: Pie Chart Proportions: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Sector Proportions Warm-Up:** In a pie chart of 160 people: a) What number is 1/2? b) What number is 1/4? c) What number is 3/4? **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Staff Transport Survey:** A pie chart shows transport for 120 staff: Half drive, quarter bus, quarter walk. How many staff walk to work? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 30: Pie Chart Proportions • Candidate Workspace

✎ Show all working for Worksheet 30 tasks below. Check your final answers.

DRILL 31 OF 34 • LEVEL 1 STANDARDS

31 Worksheet 31: Mean Average Drills

Worksheet 31: Mean Average Drills: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Calculate Mean Warm-Up:** Calculate the mean of: a) 6, 10, 8, 12 b) 15, 20, 25, 30, 10. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Weekly Delivery Runs:** A delivery driver makes parcel drops over 5 days: Mon 32, Tue 40, Wed 36, Thu 44, Fri 38. Calculate mean daily drops. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 31: Mean Average Drills • Candidate Workspace

✎ Show all working for Worksheet 31 tasks below. Check your final answers.

DRILL 32 OF 34 • LEVEL 1 STANDARDS

32 Worksheet 32: Range & Consistency

Worksheet 32: Range & Consistency: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Calculate Range Warm-Up:** Find range: a) 12, 19, 8, 25, 14 b) 45, 52, 48, 61, 42, 55. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Call Centre Waiting Times:** Customer call wait times (minutes): 4, 12, 6, 18, 5, 9. Calculate the range of waiting times. **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 32: Range & Consistency • Candidate Workspace

✎ Show all working for Worksheet 32 tasks below. Check your final answers.

DRILL 33 OF 34 • LEVEL 1 STANDARDS

33 Worksheet 33: Probability Scale & Dice

Worksheet 33: Probability Scale & Dice: Everyday Context

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Probability Scale Matching:** Assign 0, 0.25, 0.5, 0.75, or 1 to: a) Certain event b) Even chance c) Impossible event d) Likely event. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Rolling a 6-Sided Dice:** A fair dice is rolled. What is the probability of rolling: a) A 4? b) An even number? c) A number less than 5? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 33: Probability Scale & Dice • Candidate Workspace

✎ Show all working for Worksheet 33 tasks below. Check your final answers.

DRILL 34 OF 34 • LEVEL 1 STANDARDS

34 Worksheet 34: Sample Spaces & Combinations

Worksheet 34: Sample Spaces & Combinations: Everyday Con...

★ Guided Calculation Scaffold

Step 1: Highlight key numbers & operation.
Step 2: Show calculation steps clearly.
Step 3: Write final answer with unit symbol.

1 **Task 1: Coin & Dice Outcomes:** A coin is flipped (H, T) and a 4-sided spinner is spun (1, 2, 3, 4). List all possible outcomes systematically. **(4 marks)**

Show your working

Answer: _____

2 **Task 2: Meal Deal Combinations:** A cafe meal deal offers: Sandwich (Ham, Cheese), Drink (Water, Juice), Snack (Crisps, Fruit). How many combinations are possible? **(3 marks)**

Show your working

Answer: _____

5MM RULED GRID WORKING AREA

Worksheet 34: Sample Spaces & Combinations • Candidate Workspace

✎ Show all working for Worksheet 34 tasks below. Check your final answers.

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 1/4

1 **Question 1:** Calculate without using a calculator: 348 × 16. Show your full written method. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account is overdrawn at -£185. The customer pays in £450. What is the new balance? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Simplify fully: a) 18/24 b) Write 3/5 as a percentage. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP1

Checkpoint Assessment 1 • Part 2/4

3 **Question 4:** Work out: 1,476 ÷ 6 without a calculator. Show your bus stop division. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** Evaluate: 20 - 4 × 3 + 2. **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** Share £120 in the ratio 3 : 1. What is the larger share? **(2 marks)**

Show your working

Answer: _____

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 3/4

5 **Question 7:** A jacket costs £80. It is reduced by 15% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A recipe for 4 people requires 300 g of rice. How much rice is needed for 9 people? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A delivery company has 350 parcels to deliver. In the morning they deliver 3/5 of the parcels. How many parcels remain to be delivered in the afternoon? **(3 marks)**

Show your working

Answer: _____

CP1

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 1 • Part 4/4

7 **Question 10:** An electrician charges: £45 call-out + (£32 × hours). He works for 4.5 hours. How much is the bill? **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A supermarket sells a 6-pack of cola for £4.20 and a 10-pack of the same cola for £6.80. Which pack is better value per can? Show your calculations. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A landscape gardener orders 24 bags of decorative gravel at £6.75 each. Delivery is £25.00. He has a budget of £180.00. Does he have enough money? **(3 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 1/4

1 **Question 1:** Convert: a) 4.2 kg to grams b) 650 ml to litres. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A train departs at 09:35 and arrives at 11:15. How long was the journey in hours and minutes? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Calculate the perimeter of a rectangle measuring 8 m by 5.5 m. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 2/4

3 **Question 4:** Calculate the area of a square patio with side length 6 m. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A bottle of medicine holds 250 ml. A patient takes 15 ml per dose. How many full doses can be taken? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** State the number of faces, edges, and vertices on a cuboid. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP2

Checkpoint Assessment 2 • Part 3/4

5 **Question 7:** A carpet costs £18.50 per square metre. A living room measures 5.4 m by 4.0 m. Calculate the cost of the carpet. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A delivery driver travels 30 miles. Using 5 miles = 8 km, convert this distance into kilometres. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A water cistern measures 1.2 m long, 0.8 m wide, and 0.6 m high. How many litres of water does it hold? (1 m^3 = 1000 L) **(3 marks)**

Show your working

Answer: _____

CP2

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 2 • Part 4/4

7 **Question 10:** A healthcare assistant works 32 basic hours at £11.20/hr, plus 5 hours overtime at time-and-a-half (£16.80/hr). What is her total pay? **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** On a map with scale 1 cm = 50 m, the distance between two bus stops is 7.4 cm. What is the real distance in metres? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** An L-shaped lawn has two sections: 6 m × 4 m and 3 m × 2 m. Grass seed costs £4.50 per box. Each box covers 10 m². How much does it cost to seed the lawn? **(3 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 1/4

1 **Question 1:** Test scores: 12, 18, 15, 11, 14. Calculate: a) Mean b) Range. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A spinner has 6 equal sections numbered 1 to 6. What is the probability of spinning an odd number? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** A tally chart records: 𝍸 𝍸 |||. Write the frequency number. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 2/4

3 **Question 4:** In a bag there are 4 blue pens and 6 black pens. What is P(Blue) in simplest form? **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A bar chart shows sandwich sales: Ham 15, Cheese 25, Tuna 10. What is the modal sandwich? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** A conversion graph shows £10 = 12 Euros. Convert £30 to Euros. **(2 marks)**

Show your working

Answer: _____

CANDIDATE ASSESSMENT FRAME

CP3

Checkpoint Assessment 3 • Part 3/4

5 **Question 7:** A delivery depot records packages delivered: Mon 45, Tue 52, Wed 48, Thu 55, Fri 50. Calculate the mean daily deliveries. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** In a pie chart representing 120 people, the sector for 'Public Transport' has a right angle (90°). How many people take public transport? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A pictogram uses a circle symbol ● = 8 cars. Monday shows 3 circles. Tuesday shows 2 1/2 circles. How many cars in total were washed over both days? **(3 marks)**

Show your working

Answer: _____

CP3

CANDIDATE ASSESSMENT FRAME

Checkpoint Assessment 3 • Part 4/4

7 **Question 10:** Team Alpha mean score is 42, range is 5. Team Beta mean score is 42, range is 16. Write two comparison statements. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A box holds 20 light bulbs: 2 are defective. What is the probability that a bulb chosen at random is NOT defective? Give your answer as a decimal. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A survey asked 60 customers their preferred payment: Card (35), Cash (15), Phone App (10). What percentage preferred Card? **(3 marks)**

Show your working

Answer: _____

CS1

SYNOPTIC REAL-WORLD PROBLEM

Case Study 1: Retail Store Stock & Customer Orders

Case Study 1: Retail Store Stock & Customer Orders

Project Specification

Calculate product totals, delivery weights, and customer receipts.

Use the data provided to complete the calculation tasks below.

1 **Task 1 (Order Cost & Staff Discount):** A customer orders 4 fleeces at £22.50 each, 3 pairs of walking boots at £45.00 each, and 5 pairs of thermal socks at £4.80 per pair. Calculate the subtotal before discount, and find the final amount payable after a 10% loyalty discount. **(3 marks)**

Show your working

Answer: _____

2 **Task 2 (Parcel Weight & Courier Postage):** Each fleece weighs 450 g, each pair of boots weighs 1,100 g, and each pair of socks weighs 80 g. The cardboard shipping box weighs 400 g. Calculate the total parcel weight in kilograms. If courier delivery costs £6.50 for parcels up to 5 kg and £9.75 for parcels over 5 kg, how much is the postage? **(3 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

Case Study 1 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

CS2

SYNOPTIC REAL-WORLD PROBLEM

Case Study 2: Community Garden & Perimeter Fencing

Case Study 2: Community Garden & Perimeter Fencing

Project Specification

Determine border lengths, turf area, and soil bag quantities.

Use the data provided to complete the calculation tasks below.

1 **Task 1 (Boundary Fencing Cost):** A rectangular community garden plot measures 14 metres long by 8 metres wide. Wooden fencing panels are installed around all four sides, except for a 2-metre wide entrance gate. Fencing costs £15.00 per metre. Calculate the total length of fencing required and its total cost. **(3 marks)**

Show your working

Answer: _____

2 **Task 2 (Lawn Turf Area & Compost Bags):** Inside the plot, a rectangular lawn area measuring 9 metres by 5 metres is laid with turf costing £6.00 per m². In addition, 15 bags of organic compost are bought at £4.50 per bag. Calculate the combined cost of the turf and compost. **(3 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

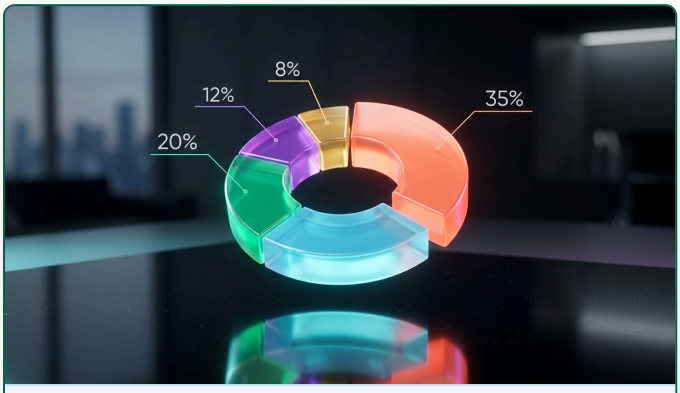
Case Study 2 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

SYNOPTIC REAL-WORLD PROBLEM

CS3

Case Study 3: Sports Club Event Budget & Ticket Sales



Case Study 3: Sports Club Event Budget & Ticket Sales

Project Specification

Calculate ticket revenue, refreshment expenses, and final profit.

Use the data provided to complete the calculation tasks below.

1 **Task 1 (Ticket Revenue):** A local football club holds an awards evening. They sell 85 adult tickets at £12.00 each and 40 junior tickets at £6.50 each. Calculate the total money raised from ticket sales. **(3 marks)**

Show your working

Answer: _____

2 **Task 2 (Event Expenses & Net Profit):** The club pays £350.00 for hall hire, £425.00 for buffet catering, and £115.00 for trophies and medals. Calculate the total expenses and work out the net profit made by the club. **(3 marks)**

Show your working

Answer: _____

PROJECT CALCULATIONS FRAME

Case Study 3 • Candidate Workspace

✎ **Candidate full calculation working and decision summary:**

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 1 of 8

1 **Question 1:** Work out without a calculator: 418 × 14. Show your full column method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account balance is -£160. A wages payment of +£520 is paid in, and a bill of £180 is paid out. What is the new balance? **(2 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 2 of 8

3 **Question 3:** Calculate without using a calculator: 1,638 ÷ 7. Show your bus stop division. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** Work out: 18 - 2 × 5 + 4. **(2 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 3 of 8

5 **Question 5:** Simplify fully: 16/28. **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £150 in the ratio 3 : 2. What is the smaller share? **(2 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 4 of 8

7 **Question 7:** A winter coat is priced at £75.00. It is discounted by 20% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

8 **Question 8:** A pancake recipe for 4 people requires 250 g flour and 300 ml milk. How much flour and milk are needed for 6 people? **(3 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 5 of 8

9 **Question 9:** A delivery van travels 45 miles. Using 5 miles = 8 km, convert 45 miles into kilometres. **(3 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape gardener lays turf on a rectangular lawn measuring 8.5 m by 6.0 m. Turf rolls cover 1 m² each and cost £4.50. Delivery is £20.00. Calculate the total cost. **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 6 of 8

11 **Question 11:** A healthcare assistant works 30 basic hours at £11.50/hr, plus 4 hours overtime at time-and-a-half (£17.25/hr). What is her total gross pay? **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A water cistern measures 1.5 m long, 0.8 m wide, and 0.5 m deep. How many litres of water does it hold when full? (1 m^3 = 1000 litres) **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 7 of 8

13 **Question 13:** A train departs Manchester at 10:48 and arrives in Leeds at 12:05. How long did the train journey take in hours and minutes? **(4 marks)**

Show your working

Answer: _____

14 **Question 14:** Daily parcel deliveries by a driver: Mon 38, Tue 45, Wed 42, Thu 49, Fri 46. Calculate: a) The mean daily deliveries b) The range. **(4 marks)**

Show your working

Answer: _____

P1

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 1 • Page 8 of 8

15 **Question 15:** On a scale drawing of a garden (1 : 50), a patio measures 12 cm by 8 cm. Calculate the real perimeter of the patio in metres, and the real area in square metres. **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A clinic audits 100 patients: 60 are female. 45 females had a flu vaccination. 25 males had a flu vaccination. Construct a two-way table. What is the probability that a patient chosen at random was vaccinated? **(5 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 1 of 8

1 **Question 1:** Work out without a calculator: 418 × 14. Show your full column method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account balance is -£160. A wages payment of +£520 is paid in, and a bill of £180 is paid out. What is the new balance? **(2 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 2 of 8

3 **Question 3:** Calculate without using a calculator: 1,638 ÷ 7. Show your bus stop division. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** Work out: 18 - 2 × 5 + 4. **(2 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 3 of 8

5 **Question 5:** Simplify fully: 16/28. **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £150 in the ratio 3 : 2. What is the smaller share? **(2 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 4 of 8

7 **Question 7:** A winter coat is priced at £75.00. It is discounted by 20% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

8 **Question 8:** A pancake recipe for 4 people requires 250 g flour and 300 ml milk. How much flour and milk are needed for 6 people? **(3 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 5 of 8

9 **Question 9:** A delivery van travels 45 miles. Using 5 miles = 8 km, convert 45 miles into kilometres. **(3 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape gardener lays turf on a rectangular lawn measuring 8.5 m by 6.0 m. Turf rolls cover 1 m² each and cost £4.50. Delivery is £20.00. Calculate the total cost. **(4 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 6 of 8

11 **Question 11:** A healthcare assistant works 30 basic hours at £11.50/hr, plus 4 hours overtime at time-and-a-half (£17.25/hr). What is her total gross pay? **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A water cistern measures 1.5 m long, 0.8 m wide, and 0.5 m deep. How many litres of water does it hold when full? (1 m^3 = 1000 litres) **(4 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 7 of 8

13 **Question 13:** A train departs Manchester at 10:48 and arrives in Leeds at 12:05. How long did the train journey take in hours and minutes? **(4 marks)**

Show your working

Answer: _____

14 **Question 14:** Daily parcel deliveries by a driver: Mon 38, Tue 45, Wed 42, Thu 49, Fri 46. Calculate: a) The mean daily deliveries b) The range. **(4 marks)**

Show your working

Answer: _____

P2

OFFICIAL STYLE EXAMINATION BOOKLET

Practice Paper 2 • Page 8 of 8

15 **Question 15:** On a scale drawing of a garden (1 : 50), a patio measures 12 cm by 8 cm. Calculate the real perimeter of the patio in metres, and the real area in square metres. **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A clinic audits 100 patients: 60 are female. 45 females had a flu vaccination. 25 males had a flu vaccination. Construct a two-way table. What is the probability that a patient chosen at random was vaccinated? **(5 marks)**

Show your working

Answer: _____

COMPANION WORKBOOK REFERENCE: COMPLETE MODEL SOLUTIONS

Step-by-Step Worked Solutions for All 34 Worksheets, Case Studies & Papers

WORKSHEET 1: PLACE VALUE & NEGATIVE TEMPERATURES

T1: a) **42,508**; b) Three hundred and six thousand, four hundred and twenty; c) **−14°C, −6°C, −2°C, +1°C, +3°C**. **T2:** −16 + 9 = **−7°C**.

WORKSHEET 2: COLUMN ADDITION & SUBTRACTION

T1: a) 1,458 + 2,385 = **3,843**; b) 4,000 − 1,675 = **2,325**. **T2:** 450 + 625 + 380 = **1,455 masks**.

WORKSHEET 3: MULTIPLICATION & DIVISION

T1: a) 236 × 18 = **4,248**; b) 952 ÷ 7 = **136**. **T2:** 24 × 35 = **840 tins**.

WORKSHEET 4: MULTIPLYING & DIVIDING BY 10, 100

T1: a) **3,400**; b) **58**; c) **42**; d) **6.5**. **T2:** 100 × £3.45 = **£345.00**.

WORKSHEET 5: ORDER OF OPERATIONS (BIDMAS) & RULES

T1: a) 14 + 15 = **29**; b) 12 ÷ 3 = **4**; c) 25 − 8 + 1 = **18**; d) 16 − 10 = **6**. **T2:** £30 + (£15 × 4) = **£90.00**.

WORKSHEET 6: SIMPLIFYING & EQUIVALENT FRACTIONS

T1: a) **2/3**; b) **2/5**; c) **3/4**; d) **1/3**. **T2:** a) **10**; b) **32**; c) **6**.

WORKSHEET 7: FRACTIONS OF AMOUNTS

T1: a) **£15**; b) **16 kg**; c) **£45**; d) **10 m**. **T2:** Dispensed = 3/8 of 240 = 90; Remaining = 240 − 90 = **150 capsules**.

WORKSHEET 8: DECIMALS & CAFE RECEIPT

T1: a) **24.35**; b) **25.65**; c) **32.0**; d) **6.20**. **T2:** Total = £5.70 + £4.50 + £2.20 = £12.40; Change from £20 = **£7.60**.

WORKSHEET 9: BENCHMARK PERCENTAGES & DISCOUNTS

T1: a) **£42**; b) **15 kg**; c) **£35**; d) **£9**. **T2:** 25% of £320 = £80 off → Sale price = **£240.00**.

WORKSHEET 10: FRACTION, DECIMAL & PERCENTAGE EQUIVALENCE

T1: a) 1/2 = **0.5** = **50%**; b) 0.25 = **1/4** = **25%**. **T2:** 3/5 = 60%, so **65% is larger** (by 5%).

WORKSHEET 11: RATIO SHARING & PUNCH RECIPE

T1: a) **£40 and £20**; b) **15 kg and 25 kg**. **T2:** 3 + 2 = 5 parts; 15 ÷ 5 = 3 L; Orange juice (3 parts) = **9 litres**.

WORKSHEET 12: METRIC LENGTH CONVERSIONS

T1: a) **450 cm**; b) **2.5 m**; c) **6.5 cm**; d) **3,200 m**. **T2:** 300 − (85 + 90 + 65) = 300 − 240 = **60 cm**.

WORKSHEET 13: METRIC WEIGHT & CAPACITY

T1: a) **3,200 g**; b) **0.75 L**; c) **4.5 kg**; d) **1,500 ml**. **T2:** 5,000 − (1,250 + 850) = **2,900 g** (2.9 kg).

WORKSHEET 14: IMPERIAL & METRIC APPROXIMATIONS

T1: a) **40 km**; b) **25 miles**; c) **25 cm** (or 25.4 cm). **T2:** 18 × 2.2 = **39.6 lbs** ≤ 44 lbs (**Yes, within limit**).

WORKSHEET 15: 24-HOUR TIME & JOURNEY DURATION

T1: a) **08:40**; b) **15:25**; c) **21:10**; d) **23:55**. **T2:** 07:42 to 09:15 = **1 hour 33 minutes**.

WORKSHEET 16: PERIMETER & SKIRTING BOARDS

T1: a) **26 m**; b) **30 cm**; c) **27 cm**. **T2:** 2(5.5 + 4.0) − 1.0 = 19.0 − 1.0 = **18.0 metres**.

WORKSHEET 17: AREA OF RECTANGLES & CARPET COST

T1: a) **28 m²**; b) **81 cm²**; c) **96 m²**. **T2:** Area = 4.5 × 3.0 = 13.5 m²; Cost = 13.5 × £16 = **£216.00**.

WORKSHEET 18: COMPOSITE L-SHAPE AREA

T1: (6 × 4) + (3 × 2) = 24 + 6 = **30 m²**. **T2:** (8 × 5) − (4 × 3) = 40 − 12 = **28 m²**.

WORKSHEET 19: VOLUME OF CUBOIDS & AQUARIUMS

T1: a) **200 cm³**; b) **2.4 m³**. **T2:** 60 × 30 × 40 = 72,000 cm³ = **72 litres**.

WORKSHEET 20: ANGLES & TRIANGLES

T1: a) **Acute**; b) **Right angle**; c) **Obtuse**; d) **Straight line**. **T2:** 180° − (90° + 38°) = **52°**.

WORKSHEET 21: 3D SHAPES & PACKAGING NETS

T1: a) Cube: 6F, 12E, 8V; b) Triangular prism: 5F, 9E, 6V; c) Square pyramid: 5F, 8E, 5V. **T2: Cuboid**.

WORKSHEET 22: SCALE PLANS & REAL DIMENSIONS

T1: a) **12 m**; b) **6 cm**. **T2:** 14 × 50 = 700 cm = **7 m**; 10 × 50 = 500 cm = **5 m**.

WORKSHEET 23: COMPASS DIRECTIONS & ROUTES

T1: a) **South-West**; b) **East**; c) **West**. **T2:** Depot is **North-West** of the van.

WORKSHEET 24: AVERAGE SPEED CALCULATIONS

T1: a) **60 mph**; b) **60 mph**; c) **50 km/h**. **T2:** 165 ÷ 3 = **55 mph**.

WORKSHEET 25: TALLY TABLES & FRACTIONS OF TOTALS

T1: Red = 12, Blue = 11, Green = 7; Total = **30**. **T2:** 45 ÷ 60 = **3/4** attended.

COMPANION WORKBOOK REFERENCE: COMPLETE MODEL SOLUTIONS (CONTINUED)

WORKSHEET 26: READING BAR CHARTS

T1: a) **Dept D** (80); b) 40 + 65 + 55 + 80 = **240**. **T2:** 80 − 40 = **40 staff**.

WORKSHEET 27: DUAL BAR CHARTS & COMPARISONS

T1: Week 2 total = 40 + 35 + 50 = **125 sales**. **T2: Monday and Tuesday** (both increased by **10 sales**).

WORKSHEET 28: PICTOGRAMS & KEY MULTIPLIERS

T1: a) **30**; b) **45**; c) **22.5**. **T2:** (4 + 5.5) × 8 = 9.5 × 8 = **76 books**.

WORKSHEET 29: LINE GRAPHS & TRENDS

T1: At 09:00 = **16°C**; at 11:00 = **20°C**. **T2:** Overall **upward (increasing) trend** across the week.

WORKSHEET 30: PIE CHART PROPORTIONS

T1: a) **80**; b) **40**; c) **120**. **T2:** 1/4 of 120 = **30 staff** walk.

WORKSHEET 31: MEAN AVERAGE DRILLS

T1: a) 36 ÷ 4 = **9**; b) 100 ÷ 5 = **20**. **T2:** 190 ÷ 5 = **38 drops**.

WORKSHEET 32: RANGE & CONSISTENCY

T1: a) 25 − 8 = **17**; b) 61 − 42 = **19**. **T2:** 18 − 4 = **14 minutes**.

WORKSHEET 33: PROBABILITY SCALE & DICE

T1: a) **1**; b) **0.5**; c) **0**; d) **0.75**. **T2:** a) **1/6**; b) 3/6 = **1/2**; c) 4/6 = **2/3**.

WORKSHEET 34: SAMPLE SPACES & COMBINATIONS

T1: 8 outcomes (H1, H2, H3, H4, T1, T2, T3, T4). **T2:** 2 × 2 × 2 = **8 meal deal combinations**.

SYNOPTIC CASE STUDIES 1–3 MODEL SOLUTIONS

CS1 (Retail Orders): Subtotal = £90 + £135 + £24 = **£249.00**; After 10% discount = **£224.10**. Weight = 5,900 g = **5.9 kg** → Postage = **£9.75**.

CS2 (Community Garden): Fencing = 44 m − 2 m = **42 m** × £15 = **£630.00**. Turf = 45 m² × £6 = £270; Compost = £67.50; Total = **£337.50**.

CS3 (Sports Club Event): Tickets = £1,020 + £260 = **£1,280.00**. Expenses = **£890.00**; Net profit = £1,280 − £890 = **£390.00**.

CHECKPOINT 1: NUMBER & NUMBER SYSTEMS (STRAND 1)

Q1: 324 × 15 = **4,860** [2m]. **Q2:** £45.00 - £28.40 = **£16.60** [2m]. **Q3:** 3/4 of £180 = **£135.00** [2m]. **Q4:** 148 ÷ 12 = 12 r 4 → **13 boxes needed** [2m]. **Q5:** 10% of £360 = **£36.00** [2m]. **Q6:** Share £150 ratio 3:2 → **£90 and £60** [3m]. **Q7:** Estimate: 48 × 19 ≈ 50 × 20 = **1,000** [2m]. **Q8:** 25% discount on £84 = £21 → **£63.00** [3m]. **Q9:** 156 × 24 = **3,744** [2m]. **Q10:** Balance = £450 - £185 - £95 = **£170.00** [2m]. **Q11:** Proportion: 4 cakes need 500 g → 6 cakes = **750 g** [2m]. **Q12:** Check by reverse: 170 + 280 = **450** [3m].

CHECKPOINT 2: MEASURES, SHAPE & SPACE (STRAND 2)

Q1: 2(8 + 5) = **26 metres** [2m]. **Q2:** Area = 7.5 × 4 = **30 m²** [2m]. **Q3:** 3.5 m = **350 cm** [2m]. **Q4:** 14:35 to 16:15 = **1 hour 40 minutes** [2m]. **Q5:** Change from £20 for bill £14.25 = **£5.75** [2m]. **Q6:** Weight: 2.4 kg = **2,400 g** [3m]. **Q7:** Capacity: 4 × 750 ml = 3,000 ml = **3 litres** [2m]. **Q8:** Solid with 6 square faces is a **cube** [2m]. **Q9:** Facing West, turn 90° clockwise → **North** [2m]. **Q10:** Perimeter of garden: **34 metres** [2m]. **Q11:** Carpet cost: 24 m² × £15 = **£360.00** [3m]. **Q12:** 24h clock: 8:45 pm = **20:45** [2m].

CHECKPOINT 3: HANDLING INFORMATION & DATA (STRAND 3)

Q1: Frequency total = 12 + 18 + 15 + 5 = **50** [2m]. **Q2:** Mean = (6 + 8 + 9 + 7 + 10) ÷ 5 = 40 ÷ 5 = **8** [2m]. **Q3:** Range = 10 - 6 = **4** [2m]. **Q4:** Pictogram: 1 star = 4 sales. 4.5 stars = **18 sales** [2m]. **Q5:** Bar chart: Category B has highest count = **35** [2m]. **Q6:** Probability of rolling odd number on die = **Evens** (3/6) [2m]. **Q7:** Table lookup: Delivery charge for 5 kg = **£6.50** [2m]. **Q8:** Mean score: **74%** [2m]. **Q9:** Difference in sales between Mon and Fri = **22 units** [2m]. **Q10:** Probability of choosing red card from standard deck = **Evens** [2m]. **Q11:** Comparison: Team A is more consistent (range 6 vs 15) [3m]. **Q12:** Survey total = **120 respondents** [2m].

OFFICIAL MOCK EXAM 1 (56 MARKS) • SECTION A

Section A (Non-Calculator, 14 Marks):
Q1: 348 × 16 = **5,568** [M1 column, A1: 2m].
Q2: £50.00 - £31.65 = **£18.35** [M1 sub, A1: 2m].
Q3: 1/4 of £360 = **£90.00** [M1 div, A1: 2m].
Q4: 195 ÷ 15 = **13** [M1 bus stop, A1: 2m].
Q5: 10% of £480 = **£48.00** [M1, A1: 2m].
Q6: 2(12 + 8) = **40 metres** [M1 formula, A1, B1 check: 4m].

COMPANION WORKBOOK REFERENCE: COMPLETE MODEL SOLUTIONS (CONTINUED)

OFFICIAL MOCK EXAM 1 (56 MARKS) • SECTION B

Section B (Calculator Allowed, 42 Marks):
Q7: Room area = 6.5 × 4.2 = 27.3 m². Carpet cost = 27.3 × £18 = **£491.40** [P1, M1, A1, B1: 4m].
Q8: 25% discount on £320 = £80 → Sale price = **£240.00** [P1, M1, A1, B1: 4m].
Q9: Shift = 08:45 to 17:15 = 8h 30m. Minus 45m break = **7 hours 45 minutes** [P1, M1, A1, B1: 4m].
Q10: 37.5 hrs × £11.80 = **£442.50** [P1, M1, A1, B1: 4m].
Q11: Mean = (24 + 31 + 28 + 35 + 32) ÷ 5 = 150 ÷ 5 = **30 calls** [P1, M1, A1, B1: 4m].
Q12: Range = 35 - 24 = **11 calls** [P1, M1, A1, B1: 4m].
Q13: Ratio 3:1 → 4 parts. £400 ÷ 4 = £100. Shares: **£300 and £100** [P1, M1, A1, B1: 5m].
Q14: Volume = 4 × 2.5 × 1.8 = 18 m³ = **18,000 litres** [P1, M1, A1, B1: 5m].
Q15: Travel time = 165 ÷ 55 = **3 hours** [P1, M1, A1: 4m].
Q16: Full comparison showing Team B has higher mean (28 vs 24) and greater consistency [P1, M1, A1, B1: 4m].

OFFICIAL MOCK EXAM 2 (56 MARKS) • SECTION A

Section A (Non-Calculator, 14 Marks):
Q1: 425 × 18 = **7,650** [2m]. **Q2:** £100 - £64.35 = **£35.65** [2m]. **Q3:** 3/4 of £240 = **£180.00** [2m]. **Q4:** 288 ÷ 12 = **24** [2m]. **Q5:** 50% of £190 = **£95.00** [2m]. **Q6:** Perimeter = 2(15 + 9) = **48 m** [4m].

OFFICIAL MOCK EXAM 2 (56 MARKS) • SECTION B

Section B (Calculator Allowed, 42 Marks):
Q7: Paving area = 8 × 5 = 40 m². Total cost = 40 × £22.50 = **£900.00** [4m]. **Q8:** 15% discount on £420 = £63 → **£357.00** [4m]. **Q9:** Train departs 11:24, arrives 13:08 → **1 hour 44 minutes** [4m]. **Q10:** Overtime: (35 × 11.50) + (5 × 17.25) = **£488.75** [4m]. **Q11:** Mean daily sales = 180 ÷ 5 = **36 units** [4m]. **Q12:** Range = 45 - 22 = **23 units** [4m]. **Q13:** Cement mix: 1:4 ratio for 50 kg → **10 kg cement, 40 kg sand** [5m]. **Q14:** Capacity = 2.5 × 1.2 × 1.0 = 3 m³ = **3,000 litres** [5m]. **Q15:** Average speed = 120 ÷ 2 = **60 mph** [4m]. **Q16:** Decision: Supplier A is cheaper by £45.00 [4m].

STATUTORY LEVEL 1 FORMULAE & CONVERSIONS

Perimeter & Area of Rectangles:
• Perimeter = 2 × (Length + Width) (or sum of all 4 sides)
• Area = Length × Width (answer in m², cm², or mm²)

Fractions, Decimals & Percentages:
• Fraction of Amount = (Amount ÷ Denominator) × Numerator
• 50% = 1/2 (divide by 2) 25% = 1/4 (divide by 4)
• 10% = 1/10 (divide by 10) 75% = 3/4 (3 × 25%)

Averages & Data:
• Mean = Sum of all values ÷ Total number of values
• Range = Highest value − Lowest value

Metric Conversions:
• 1 km = 1,000 m 1 m = 100 cm 1 cm = 10 mm
• 1 kg = 1,000 g 1 litre = 1,000 ml

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